CS577

Principles and Techniques of Data Science

Phase 1

Dataset & Research Proposal

Define your problem

Overview

This phase is about defining your problem clearly and proving that your dataset is appropriate for analysis. You are not expected to build models yet. You are expected to think carefully.

Strong foundations now will make the rest of the project significantly easier.

Submission Requirements

Submit a single PDF (*GroupNumber_FirstName1_FirstName2_Phase1.pdf*) containing all sections listed below. You must also attach your dataset file (CSV) or provide a direct download link.

Section 1: Title

Provide a short, specific title. Your title must clearly indicate:

- The main variable of interest
- The population or dataset being studied

Example:

"Predicting 30-Day Hospital Readmission Using Medicare Data"

Section 2: Research Question

Your research question should follow one of these structures:

"Does X predict Y in population Z?"

"How does X vary across Y over time?"

No vague topics. Write 1 to 3 clear sentences answering:

- What exactly are you trying to investigate?
- Is your project descriptive, predictive, or explanatory?

Required Sentence

> *One row in my dataset represents _____.*

This sentence is required. If it is missing, points will be deducted.

Section 3: Dataset Information

You must include all of the following:

- Dataset source (URL)

- How the data was collected (if known)
- License or access type (public, API, Kaggle, etc.)
- Number of rows
- Number of columns
- Data types (numeric, categorical, text, date)
- Time range (if applicable)
- Known limitations of the dataset

You must either:

- Upload the dataset file (.csv), or
- Provide a direct download link

Both options are acceptable. A dataset that cannot be accessed will result in an incomplete submission.

Section 4: Planned EDA Questions

List at least four specific exploratory questions you will answer in Phase 2. These should be concrete and tied to your dataset.

Examples:

- What is the distribution of the target variable?
- Are there missing values? In which columns and what percentage?
- Are there outliers in key numeric features?
- Are any features strongly correlated with the target?

Generic questions like "Is the data clean?" will not receive full credit. Be specific.

Section 5: Anticipated Challenges

Briefly describe at least three potential challenges you expect to encounter. Each challenge should be specific to your dataset and research question.

Examples of valid challenges:

- Missing data in key columns and a plan for handling it
- Small sample size and what that means for model generalization
- Class imbalance in the target variable
- Ethical concerns around collecting or using this data
- Feature engineering complexity (e.g., encoding high-cardinality categoricals)

"The data might be messy" is not a specific challenge. Name the column, describe the problem, explain why it matters.

Dataset Requirements

Your dataset must meet all of the following criteria:

Requirement	Details
Minimum size	1,000 rows unless explicitly justified in your proposal
Feature count	At least 5 meaningful features (not counting the row index)
Tooling	Must be analyzable using pandas in Python
No toy datasets	MNIST, Iris, Titanic, and similar tutorial datasets are not permitted
Unique topics	No two students may use the same dataset or research question

Helpful Dataset Sources

You may search independently. These are reliable starting points:

Source	URL
Kaggle Datasets	kaggle.com/datasets
UCI Machine Learning Repo	archive.ics.uci.edu
Data.gov	data.gov
FiveThirtyEight Data	github.com/fivethirtyeight/data
World Bank Open Data	data.worldbank.org
CDC Open Data	data.cdc.gov
NOAA Climate Data	climate.gov/maps-data
Our World in Data	ourworldindata.org
Bureau of Labor Statistics	bls.gov/data
NYC Open Data	opendata.cityofnewyork.us
California Open Data Portal	data.ca.gov

Grading Criteria

Your proposal will be evaluated on the following criteria:

Component	Points
Section 1: Title (clear, specific, meets format requirements)	10
Section 2: Research question (clear, typed, required sentence included)	20
Section 3: Dataset information (all required fields present, file accessible)	25
Section 4: Planned EDA questions (at least 4, specific and dataset-tied)	20
Section 5: Anticipated challenges (at least 3, specific and reasoned)	15
Dataset requirements met (size, features, no toy datasets, unique topic)	10
Total	**100**

Final Note

Running a model is not the goal of Phase 1.
The goal is to define a problem worth solving and prove you have data that can solve it. Vague questions produce vague models. Weak datasets produce weak results. The time you invest here pays dividends across every phase that follows.

If you cannot explain what one row in your dataset represents, you do not yet understand your data.